Tang MEGA 138K Pro

Specification v1.0



Release Note

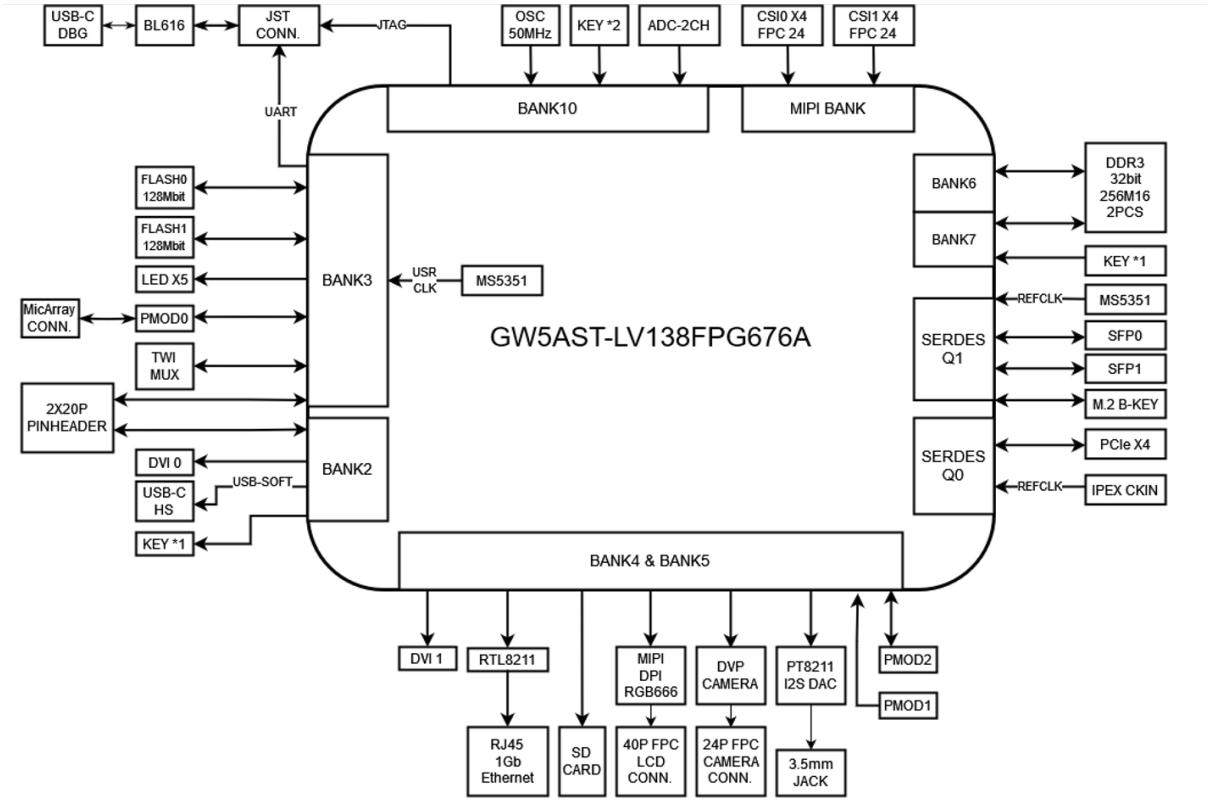
Version	Update content
V1.0	Edited on September 21, 2023; Original Version; PCB version: v4071

1. Overview

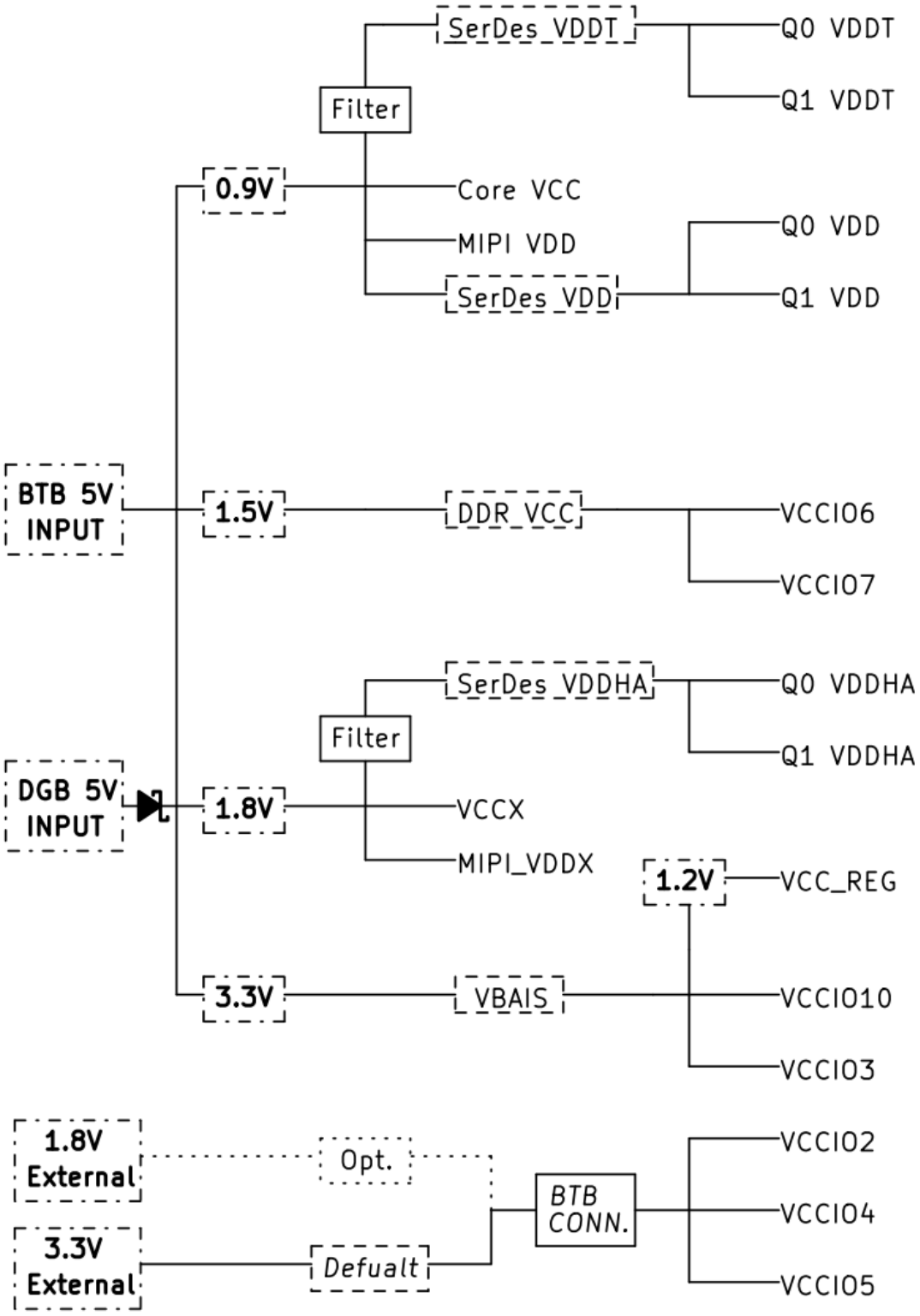
Tang MEGA 138K Pro SOM is a highly integrated FPGA SOM with a compact PCB size of only 70mm*50mm. It integrates the Gowin GW5AT-LV138FPG676A as the main chip, dual 128Mbit SPI NOR FLASH, dual DDR3 chips, each with a capacity of 4Gbit. It also has DC-DC power supply, both JST 8P debug connectors and triple 120P BTB connectors. The SOM provides 187 GPIOs, 2 hard-core 4-lane MIPI D-PHY RX, dual 12.5Gbps SERDES with 4 lane TX/RX of each. and PCIe 2.0 hard core. By simply providing the SOM with 5V power and JTAG whether via JST 8P connector or BTB connector, customers can easily use the *Tang MEGA 138K Pro SOM* to work with their ideas.

Tang MEGA 138K Pro SOM and Gowin FPGA development platform jointly help customers quickly achieve commercial landing in fields such as parallel data communication, High -speed data transmission, image processing, digital logic circuit prototype verification, and so on.

1.1 Block Diagram



1.2 Power Tree



1.3 Hardware Overview

Component	Characteristic
Tang MEGA 138K Pro SOM Note: Only the key features of the FPGA are mentioned on the right side. Please refer to the chip datasheet for detailed	Logic Units (LUT4): 138,240
	Register (REG): 138 ,240
	Distributed Static Random Access Memory SSRAM: 1,080 kbits
	Block Static Random Access Memory BSRAM: 6,120 kbits
	Number of BSRAMs: 340
	DSP (27-bit x 18-bit): 298
	Phase locked loop (PLLs): 12
	Global Clocks: 16
	High-speed Clocks: 24
	Transceivers (270Mbps - 12.5Gbps): 8
	PCIe 2.0 (Hard core): 1, Could be configured as PCIe x1, x2, x4, x8
	DDR3: 2x 4Gbit, up to 1066MT/s
	LVDS: 1.25Gbps
	MIPI D-PHY Hard core: 2.5Gbps (RX), 8 data lanes, 2 clock lanes
	ADC: 2
	Hardcore SoC: AE350, RSIC-V 32bit
	Number of GPIO banks: 6, Available IO quantity: 187
	Dimension: 70.0mm x 50.0mm
	BTB male connector in SOM: DF40C-120DP-0.4V (51) BTB female connector in Base Board: DF40C-120Ds-0.4V (51)e

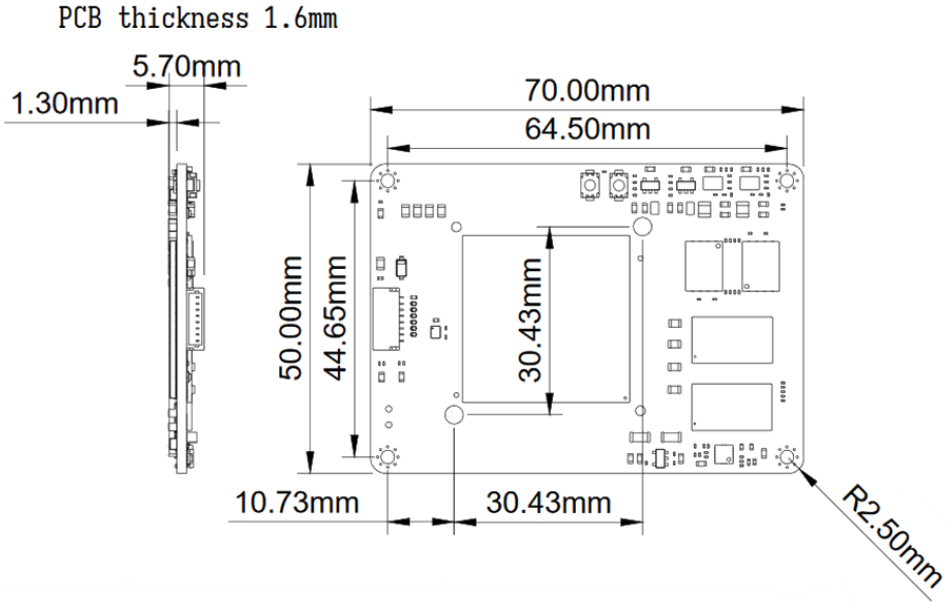
1.4 Working Conditions

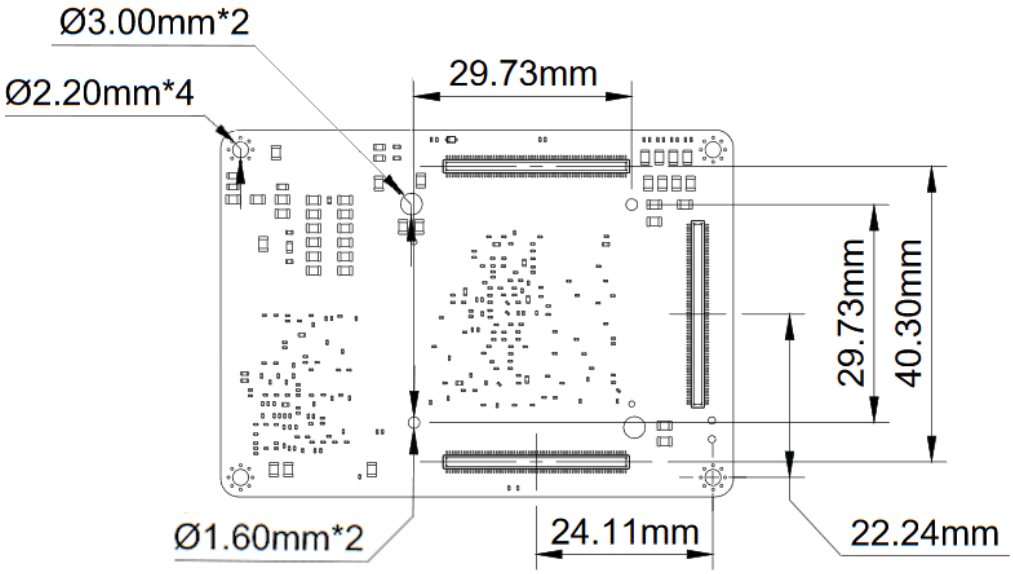
Items	Description
SOM power supply	Working Voltage: Mainboard provide 3.6~5.5V to SOM through 8P debug connector or BTB connector. Working Current: Depends on actual operating conditions, 5V@110mA ~ 1500mA in general.
Temperature rise	< 40K
Recommended operating temperature range	0℃ ~ 85℃ This series of chips has models with a wider operating temperature range to choose from. If you need it, please contact us via email
Minimum System Requirements	1. 8P debug connector provides 5V for SOM. (Main power supply) 2. Connect JTAG via 8P debug connector. (Debug the SOM via JTAG) 3. Dock board provides VCCIO2/4/5 SOM. (Bank2/4/5 power supply, not necessary)

2. Apperance Parameters

2.1 Dimension

Items	Description
Length	70.0 mm
width	50.0 mm
Thickness	Please review the 3D drawings

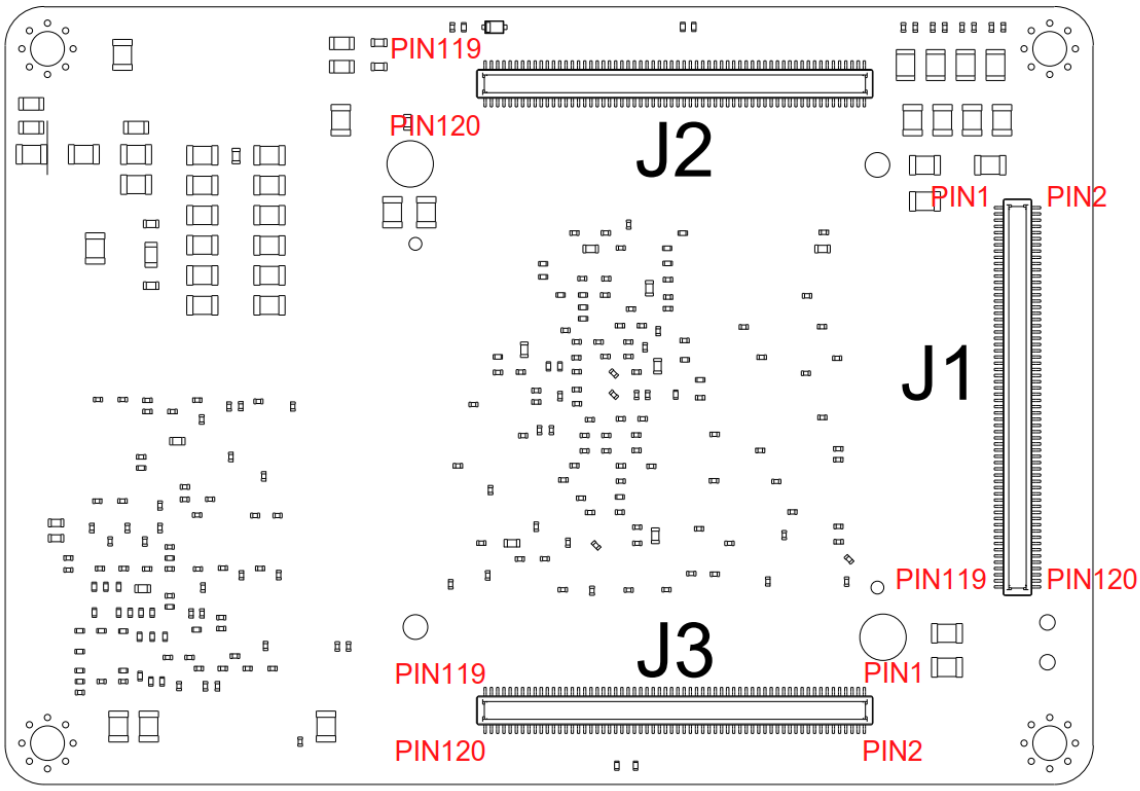




3. Description of pinout

Please refer to the document *Tang_MEGA_138K_Pro_Pinout_Track_Length_Table.xlsx* to get a detailed pinout.

3.1 SOM Bottom View of BTB Conn.



4. Notice

Item	Description
ESD protection	Please pay attention to avoid ESD hitting the PCBA. Please discharge the human static electricity before touching PCBA
IO voltage	Please don't let the actual working voltage of IO exceed the rated value, otherwise it will cause permanent damage to PCBA
Avoid short circuit	Please avoid any liquid or metal touching the pads of components on PCBA during power on, otherwise it will cause short circuit and damage the PCBA
Removing SOM	Please pull out the SOM from both sides at the same time, otherwise there is a high probability of damaging the BTB connector. If conditions permit, please use dedicated IC puller
Installing SOM	Before installing the SOM , please check that the appearance of the BTB connector of the SOM and the base board are normal. If there is any deviation, it needs to be aligned before installation. If forcibly installed, it may cause damage to the BTB connector

5. Resources

Item	Description
Official website	www.sipeed.com
Github	https://github.com/Sipeed
BBS	http://bbs.sipeed.com
Wiki	wiki.sipeed.com
Sipeed model platform	https://maixhub.com/
SDK /HDK	https://dl.sipeed.com/
E-mail (For Technical support & Business cooperation)	support@sipeed.com

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